Lab 7: Fuzzing for Input Validation Bugs

Software Engineering — Repository: L6_L7_65_solar_panel_estimate (LAB7)

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Section A

Tools Python 3.13.7 (Windows), Hypothesis 6.170.14, pytest 9.1.1

Bugs found by fuzzing and how they were fixed

Function	Failing input (Hypothesis)	Crash	Fix
`clean_site_name`	`None`	AttributeError: 'NoneType' has no attribute 'strip'	Return `None` for non-strings; strip after removing '#'/'@'; return `None` if nothing is left; cap length at 100 chars.
`parse_monthly_units`	`''`, `None`	ValueError: invalid literal for int(); AttributeError on `.lower()`	Type check, then a full-match ASCII regex `(\d{1,7})\s*(kwh)?`, case-insensitive. Also rejects `-5`, `1_000`, Unicode digits like `٣٢٠` and huge digit strings, which `int()` would accept or choke on.
`panels_for_roof`	`''`, `None`	ValueError on empty part; AttributeError. Input without an 'x' (e.g. `'0'`) also raises IndexError.	Type check, then a full-match ASCII regex for `L[ft] x W[ft]` (1–5 digits each); zero dimensions return `None`.

All three functions now return `None` for invalid input instead of raising, and `main()` prints a clear "invalid input" message. Each test asserts the function never raises and that the result is either `None` or the right type (a non-negative `int` for the two numeric parsers), over 500 generated inputs per test.

Inputs given for each run

Screenshot	Module	Site name	Monthly use	Roof size	Result
SS1(a)	Buggy	`sunny#side @farm`	`320 kWh`	`20ft x 15ft`	Sunnyside Farm, 320, 16 panels
SS1(b)	Buggy	`green valley`	`320 units`	(not reached)	Crashes with ValueError
SS2	Buggy	Generated by Hypothesis: 500 random strings + `None` per test. Minimal failing inputs found: `None` (all three), `''` (parse_monthly_units, panels_for_roof)			3 failed
SS3(a)	Fixed	`sunny#side @farm`	`320 kWh`	`20ft x 15ft`	Sunnyside Farm, 320, 16 panels
SS3(b)	Fixed	`#@#`	`320 units`	`20ft by 15ft`	"invalid input" for all three, no crash
SS4	Fixed	Same Hypothesis strategies as SS2			3 passed

SS1 — Running the buggy module: `python solar_panel_estimate_parser.py`

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python solar_panel_estimate_parse
r.py
==== Running clean_site_name ====
Enter the installation site name: sunny#side @farm
Site name: Sunnyside Farm

==== Running parse_monthly_units ====
Enter monthly electricity use (e.g. 320 kWh): 320 kWh
Monthly units: 320

==== Running panels_for_roof ====
Enter roof size as LENGTHft x WIDTHft (e.g. 20ft x 15ft): 20ft x 15ft
Panels that fit: 16
```

SS1(a): well-formed input works

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python solar_panel_estimate_parse
r.py
==== Running clean_site_name ====
Enter the installation site name: green valley
Site name: Green Valley

==== Running parse_monthly_units ====
Enter monthly electricity use (e.g. 320 kWh): 320 units
Traceback (most recent call last):
  File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimate_pars
er.py", line 40, in <module>
    main()
    ~~~~^^
  File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimate_pars
er.py", line 33, in main
    print("Monthly units:", parse_monthly_units(value))
                            ~~~~~~~~~~~~~~~~~~~~^^^^^^^
  File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimate_pars
er.py", line 16, in parse_monthly_units
    return int(text.lower().replace('kwh', '').strip())
ValueError: invalid literal for int() with base 10: '320 units'
```

SS1(b): '320 units' crashes parse_monthly_units with ValueError

SS2 — Hypothesis tests failing on the buggy module: `python -m pytest test_solar_panel_estimate_parser.py`

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python -m pytest test_solar_panel
_estimate_parser.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7
plugins: anyio-4.11.0, hypothesis-6.170.14, langsmith-0.10.1
collected 3 items

test_solar_panel_estimate_parser.py FFF                                                                                        [100%]

================================================================= FAILURES ==================================================================
_____________________________________________ test_clean_site_name_no_crash _____________________________________________

s = None

    @settings(max_examples=500)
    @given(st.text() | st.none())
    def test_clean_site_name_no_crash(s):
        try:
>           result = clean_site_name(s)
                     ^^^^^^^^^^^^^^^^^^

test_solar_panel_estimate_parser.py:9:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

name = None

    def clean_site_name(name):
        """
        Trims a site name, removes '#' and '@', and title-cases it.
        Assumes name is a non-empty string.
```

SS2 (1/6): 3 tests collected, FFF; clean_site_name traceback begins

```
    def clean_site_name(name):
        """
        Trims a site name, removes '#' and '@', and title-cases it.
        Assumes name is a non-empty string.
        """
>       name = name.strip()
               ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

solar_panel_estimate_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @settings(max_examples=500)
>   @given(st.text() | st.none())
                   ^^^

test_solar_panel_estimate_parser.py:6:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @settings(max_examples=500)
    @given(st.text() | st.none())
    def test_clean_site_name_no_crash(s):
        try:
            result = clean_site_name(s)
        except Exception as e:
>           assert False, f"clean_site_name crashed on {s!r} with: {type(e).__name__}: {e}"
E           AssertionError: clean_site_name crashed on None with: AttributeError: 'NoneType' object has no attribute 'strip'
E           assert False
```

SS2 (2/6): clean_site_name fails on None (AttributeError)

```
E           AssertionError: clean_site_name crashed on None with: AttributeError: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_clean_site_name_no_crash(
E               s=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_pa
nel_estimate_parser.py:10

test_solar_panel_estimate_parser.py:11: AssertionError
_____________________________________________ test_parse_monthly_units_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_esti
mate_parser.py", line 19, in test_parse_monthly_units_no_crash
  |     @given(st.text() | st.none())
  |                        ^^^
  |   File "C:\Users\ravik\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 22, in test_parse_monthly_units_no_crash
    |     result = parse_monthly_units(s)
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimat
e_parser.py", line 16, in parse_monthly_units
    |     return int(text.lower().replace('kwh', '').strip())
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
```

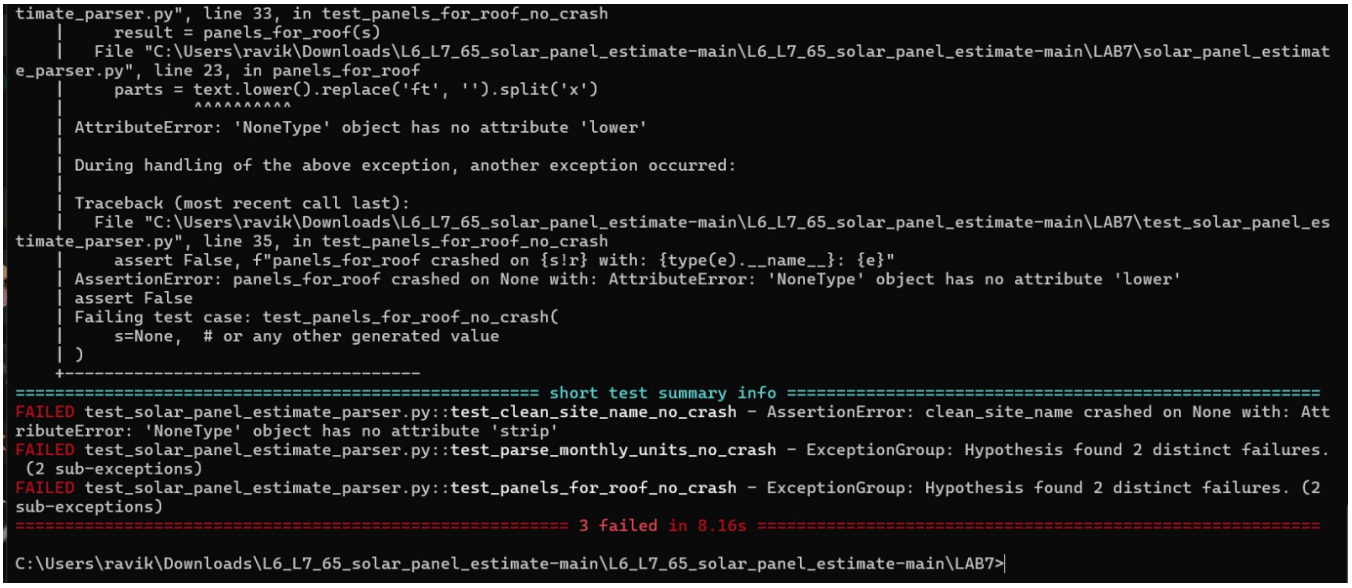
SS2 (3/6): clean_site_name failing case s=None; parse_monthly_units fails on '' (ValueError)

```
    |     return int(text.lower().replace('kwh', '').strip())
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 24, in test_parse_monthly_units_no_crash
    |     assert False, f"parse_monthly_units crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_monthly_units crashed on '' with: ValueError: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_monthly_units_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 22, in test_parse_monthly_units_no_crash
    |     result = parse_monthly_units(s)
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimat
e_parser.py", line 16, in parse_monthly_units
    |     return int(text.lower().replace('kwh', '').strip())
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'lower'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 24, in test_parse_monthly_units_no_crash
    |     assert False, f"parse_monthly_units crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: parse_monthly_units crashed on None with: AttributeError: 'NoneType' object has no attribute 'lower'
```

SS2 (4/6): parse_monthly_units fails on '' and on None (AttributeError)

```
_____________________________________________ test_panels_for_roof_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_esti
mate_parser.py", line 30, in test_panels_for_roof_no_crash
  |     @given(st.text() | st.none())
  |                        ^^^
  |   File "C:\Users\ravik\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2330, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 33, in test_panels_for_roof_no_crash
    |     result = panels_for_roof(s)
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\solar_panel_estimat
e_parser.py", line 24, in panels_for_roof
    |     return int(parts[0]) * int(parts[1]) // 18
    |            ~~~^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7\test_solar_panel_es
timate_parser.py", line 35, in test_panels_for_roof_no_crash
    |     assert False, f"panels_for_roof crashed on {s!r} with: {type(e).__name__}: {e}"
    | AssertionError: panels_for_roof crashed on '' with: ValueError: invalid literal for int() with base 10: ''
```

SS2 (5/6): panels_for_roof fails on '' (ValueError)



SS2 (6/6): panels_for_roof fails on None (AttributeError); summary: 3 failed

SS3 — Running the fixed module: `python solar_panel_estimate_parser.py`

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python solar_panel_estimate_parse
r.py
==== Running clean_site_name ====
Enter the installation site name: sunny#side @farm
Site name: Sunnyside Farm

==== Running parse_monthly_units ====
Enter monthly electricity use (e.g. 320 kWh): 320 kWh
Monthly units: 320

==== Running panels_for_roof ====
Enter roof size as LENGTHft x WIDTHft (e.g. 20ft x 15ft): 20ft x 15ft
Panels that fit: 16

C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>
```

SS3(a): well-formed input gives the same results as before

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python solar_panel_estimate_parse
r.py
==== Running clean_site_name ====
Enter the installation site name: #@#
Site name: invalid input (name cannot be empty)

==== Running parse_monthly_units ====
Enter monthly electricity use (e.g. 320 kWh): 320 units
Monthly units: invalid input (expected e.g. 320 kWh)

==== Running panels_for_roof ====
Enter roof size as LENGTHft x WIDTHft (e.g. 20ft x 15ft): 20ft by 15ft
Panels that fit: invalid input (expected e.g. 20ft x 15ft)
```

SS3(b): malformed input is reported instead of crashing

SS4 — Hypothesis tests passing on the fixed module: `python -m pytest test_solar_panel_estimate_parser.py -v`

```
C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7>python -m pytest test_solar_panel
_estimate_parser.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.7, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\ravik\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\ravik\Downloads\L6_L7_65_solar_panel_estimate-main\L6_L7_65_solar_panel_estimate-main\LAB7
plugins: anyio-4.11.0, hypothesis-6.170.14, langsmith-0.10.1
collected 3 items

test_solar_panel_estimate_parser.py::test_clean_site_name_no_crash PASSED                                                          [ 33%]
test_solar_panel_estimate_parser.py::test_parse_monthly_units_no_crash PASSED                                                      [ 66%]
test_solar_panel_estimate_parser.py::test_panels_for_roof_no_crash PASSED                                                          [100%]

============================================================ 3 passed in 10.88s =============================================================
```

SS4: all 3 tests pass